

10.A

Properties of Logarithms

 Tarsia Puzzle

Tarsia Puzzle

Setup: Groups of 4. Each group gets one handout and a pair of scissors. Cut out all the pieces before you begin.

1. **Examine the Pieces:** Each piece has expressions, answers, or steps along its edges. Your job is to match edges that belong together.
2. **Build the Puzzle:** Place pieces so that matching edges touch. When all pieces are correctly matched, they will form a complete shape.
3. **Check Your Shape:** If the pieces form a clean shape with no gaps or overlaps, your matches are correct. If anything doesn't fit, there is an error somewhere — find it and fix it.
4. **Troubleshoot Together:** If the shape doesn't close, go back and check each match. Work the math, don't just try random placements.
5. **When Finished:** Raise your hand for the teacher to verify, then discuss: were there any matches that surprised you?

✓ Looks Like Success

- Every match is verified by working the math, not guessing
- Group catches a wrong match by checking the shape, not asking the teacher
- All four members can explain why each edge pair belongs together

✗ What to Avoid

- Forcing pieces to fit without checking the math
- One person assembling while others watch
- Calling it done before the shape is fully closed and clean







